FairChild Institute

Data Analysis Institute

12-Month Practical Data Analysis Mentorship Program

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Month 1

Foundations of Data Analytics & Excel Essentials

Phase: FOUNDATION | Theme: From Spreadsheet User to Data Thinker

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Data Analytics & Spreadsheet Discipline	How should an analyst think before touching the data?	['Audit a deliberately messy sale workbook and produce a short data-quality log.', 'Create a clean analysis-ready workbook structure and explain every design choice.']
2	Formulas, Logic & Lookups	How do formulas turn raw records into useful information?	['Build a customer/order analysis model using formulas and lookups.', 'Diagnose and repair formula errors in an unfamiliar workbook.']
3	Cleaning, Validation & Quality Control	How do I know the dataset and calculations can be trusted?	['Clean a workforce dataset and document every material transformation.', 'Reconcile pre-cleaning and post-cleaning totals and defend the cleaning decisions.']
4	Pivot Analysis & Executive Summary	How can I move from rows of data to a defensible insight?	['Analyze a retail dataset and create a one-page Excel management summary.', 'Present three evidence-based insights, one limitation and one recommended action.']

Final Project	Excel Business Performance Analysis

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Month 2

Advanced Excel Modelling, Power Query & Dashboarding

Phase: FOUNDATION | Theme: From Analysis to Repeatable Workflow

Duration 4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Advanced Functions & Dynamic Analysis	How can Excel models remain flexible as data changes?	['Build a dynamic sale performance analysis that updates with new records.', 'Replace brittle manual calculations with a maintainable formula design.']
2	Power Query Fundamentals	How can repetitive cleaning become a refreshable process?	['Transform monthly raw files into one clean table using Power Query.', 'Explain the query steps and prove the refresh works with a new file.']
3	Excel Modelling & Controls	How should a professional analytical workbook be designed?	['Build a budget-versus-actual model with scenarios and control checks.', 'Find and fix hidden model risks in a provided workbook.']
4	Excel Dashboard & Month Project	What makes a dashboard useful rather than merely attractive?	['Create a refreshable executive dashboard from Power Query output.', 'Deliver a five-minute management briefing without walking through every worksheet.']

Final Project Refreshable Excel Operations Dashboard

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Month 3

SQL Foundations & Relational Data

Phase: QUERY | Theme: From Workbook Tables to Database

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Relational Thinking & SELECT	How is database data organized differently from a spreadsheet?	'Explore a training database and answer structured business questions.', 'Write readable queries and explain the grain of each result.']
2	Filtering, Logic & Functions	How do I retrieve exactly the records I need?	['Create customer and transaction segments from business rules.', 'Translate ambiguous business rules into explicit query conditions.']
3	Aggregation & Grouped Analysis	How do I summaries millions of rows without exporting them?	['Produce a regional performance report entirely in SQL.', 'Validate grouped totals against base-table totals.']
4	Joins & Relational Analysis	How do separate tables become one analytical view?	['Combine customers, orders and products to answer profitability questions.', 'Diagnose a deliberately incorrect join and prove the corrected result.']

Final Project	SQL Customer & Sales Analysis

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Month 4

Intermediate SQL & Analytical Query Design

Phase: QUERY | Theme: From Query Writer to SQL Analyst

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Subqueries & CTEs	How can I break a complex SQL problem into understandable stages?	['Build a multi-stage profitability analysis using CTEs.', 'Refactor an unreadable query into auditable analytical steps.']
2	Window Functions	How can I compare each row with its group or history?	['Rank products and analyze month-over-month change without collapsing detail.', 'Explain why a window function is preferable to GROUP BY for selected tasks.']
3	Time, Cohorts & Retention	How do analysts measure behavior over time?	['Create a customer cohort/ retention table from transaction history.', 'Investigate an unexpected retention pattern and document the evidence.']
4	Reusable SQL & Performance Awareness	How do I make analytical SQL maintainable and efficient?	['Build a reusable analytical view and a management query pack.', "Peer-review another learner's SQL for correctness, clarity and efficiency."]

Final Project	SQL Retention & Performance Case Study

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Month 5

Python for Data Analytics

Phase: PROGRAMMATIC ANALYSIS | Theme: From Manual Analysis to Reproducible Code

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Python & Notebook Foundations	How does code make analysis reproducible?	['Recreate a simple spreadsheet analysis in Python.', 'Write a small reusable function and explain each step.']
2	pandas: Insp	How can I manipulate tabular data reliably in code?	['Clean a messy CSV and produce a data-quality report.', 'Prove that cleaning did not accidentally change key totals.']
3	Group, Merge & Reshape	How do I combine and summarize datasets programmatically?	['Combine transaction and reference data to build an analytical table.', 'Diagnose a many-to-many merge problem.']
4	EDA & Python Reporting	How can code reveal patterns without encouraging false conclusions?	['Combine customers, orders and products to answer profitability questions.', 'Diagnose a deliberately incorrect join and prove the corrected result.']

Final Project	Python Exploratory Data Analysis

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Month 6

R for Statistical Analysis & Analytical Reasoning

Phase: PROGRAMMATIC ANALYSIS | Theme: From Description to Statistical Evidence

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	R & Tidy Data Foundations	How does R support reproducible statistical analysis?	['Reproduce a descriptive analysis in R.', 'Compare an R workflow with the equivalent Python/Excel approach.']
2	Visualization & Distribution Thinking	What does the shape of the data tell us?	['Create an exploratory visual report for a public/business dataset.', 'Explain what each chart shows and what it cannot establish.']
3	Statistical Inference for Analysts	When is an observed difference meaningful?	['Evaluate a realistic A/B or group comparison question.', "Write a non-technical interpretation without saying 'proved'"]
4	Correlation, Regression & Responsible Interpretation	How can we model relationships without confusing association with causation?	['Fit and interpret a simple regression model.', 'Challenge an overconfident causal claim using analytical evidence.']

Final Project	R Statistical Evidence Report

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Month 7

Power BI Foundations & Data Modelling

Phase: BUSINESS INTELLIGENCE | Theme: From Analysis to Interactive BI

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Power BI Workflow & Power Query	How does raw source data become a BI model?	['Prepare a multi-table business dataset for modelling.', 'Document every material transformation and refresh dependency.']
2	Data Modelling	Why does model design determine whether a dashboard can be trusted?	['Design a sales model from raw operational tables.', 'Diagnose ambiguous/incorrect relationships in a broken model.']
3	DAX Foundations	How do measures respond to the user's filter context?	['Create a KPI measure library and validate each measure.', 'Explain why a total differs from a naive row-level calculation.']
4	Interactive Reporting	How should users move from overview to explanation?	['Build a two-page interactive management report.', 'Conduct a usability review with another learner and revise the report.']

Final Project	Power BI Sales & Operations Report

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Month 8

Advanced Power BI, DAX & Dashboard Design

Phase: BUSINESS INTELLIGENCE | Theme: From Dashboard Builder to BI Analyst

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Advanced DAX & Filter Context	How can I control what a measure calculates?	['Build margin, share-of-total and benchmark measures.', 'Debug a set of intentionally incorrect DAX measures.']
2	Time Intelligence & Variance	How do we compare performance across time fairly?	['Create an executive time-performance measure pack.', 'Explain calendar assumptions and test incomplete-period behaviour.']
3	Advanced UX & Analytical Navigation	How can dashboard interaction support investigation rather than distraction?	['Redesign a cluttered dashboard around a clear decision journey.', 'Run a stakeholder-style usability test and log improvements.']
4	Performance, Governance & BI Review	How can a BI solution remain trustworthy after handover?	['Create a BI handover pack and final QA checklist.', 'Peer-review another solution for accuracy, usability and maintainability.']

Final Project	Executive Power BI Performance Cockpit

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Month 9

Integrated Analytics: SQL + Python/R + Power BI

Phase: INTEGRATION | Theme: From Tool User to Analytics Workflow Designer

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Problem Framing & Analytical Architecture	What should happen in SQL, code and BI - and why?	['Design an architecture for an end-to-end analytics case.', 'Defend tool choices and identify what should not be moved into Power BI.']
2	SQL Extraction & Data Preparation	How do I create a reliable analytical dataset at source?	['Build a documented SQL extraction layer.', 'Prove row counts and totals reconcile to source expectations.']
3	Python/R Analytical Layer	When does programmatic analysis add value beyond SQL?	['Create enriched analytical outputs for BI consumption.', 'Compare two possible implementations and justify the chosen approach.']
4	Power BI Delivery & Cross-Tool QA	How do I ensure the final dashboard still reflects the source correctly?	['Deliver the integrated dashboard and QA pack.', 'Trace one KPI all the way from dashboard back to source records.']

Final Project	Power BI Sales & Operations Report

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Month 10

Power BI Delivery & Cross-Tool QA

Phase: COMMUNICATION | Theme: From Correct Analysis to Influential Insight

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Visualisation Principles	What makes a chart truthful, clear and useful?	['Redesign a set of poor charts and explain each correction.', 'Create three different visuals from the same data for three different questions.']
2	Insight & Narrative	How do I move from 'what happened' to 'what should we do'?	['Turn an analysis pack into a one-page executive story.', 'Remove unsupported claims and strengthen evidence links.']
3	Documentation & Reproducibility	Could another analyst understand, reproduce and maintain my work?	['Create a professional documentation pack for a prior project.', 'Conduct a handover test: another learner must reproduce one result.']
4	Presentation & Analytical Viva	Can I communicate the same analysis to technical and non-technical audiences?	['Deliver a 7-minute executive briefing and technical Q&A.', 'Revise the story based on audience feedback.']

Final Project	Executive Analytics Story & Documentation Pack

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Month 11

Portfolio Analytics & Real-World Casework

Phase: PROFESSIONAL PRACTICE | Theme: From Learner to Junior Analyst

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Case 1: Commercial / Sales Analytics	How would an analyst investigate performance without a step-by-step brief?	['Complete an open-ended commercial analytics case.', 'Defend the KPI definitions and prioritise recommendations.']
2	Case 2: Operations / Service Analytics	How do we identify bottlenecks and performance opportunities?	['Complete an operations analytics case using self-selected tools.', 'Identify what additional data would materially improve the conclusion.']
3	Portfolio Engineering & Peer Review	What makes an analytics project credible to another professional?	['Rebuild one earlier project to professional portfolio standard.', 'Perform a formal peer review using an accuracy/clarity/reproducibility rubric.']
4	Analyst Interview & Case Defence	Can I explain my thinking under questioning?	['Complete a timed analytics case and viva.', 'Produce a personal gap-remediation plan before capstone month.']

Final Project	Professional Analytics Portfolio Release

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Month 12

Capstone: End-to-End Data Analytics Solution

Phase: CAPSTONE | Theme: From Junior Analyst to Independent Problem Solver

Duration	4 Core Instructional Weeks

Month Journey

Week	Theme	Core Question	Weekly Challenge
1	Capstone Discovery & Proposal	What problem is worth solving, and what evidence would answer it?	['Submit capstone proposal, data audit and analytical plan.', 'Defend scope and revise based on mentor review.']
2	Build Sprint: Data & Analysis	Can I independently build the analytical core?	['Produce a working analytical pipeline and interim findings.', 'Complete a checkpoint without step-by-step mentor instructions.']
3	Build Sprint: BI, Story & Documentation	Can I turn the analysis into a usable decision-support product?	['Complete the solution and documentation pack.', 'Run an independent QA and user test, then fix identified issues.']
4	Final Presentation, Viva & Reflection	Can I defend the entire analytical journey?	['Deliver final capstone presentation and viva.', 'Respond to an unfamiliar change request and explain how the solution would be adapted.']

Final Project	Independent Data Analytics Capstone